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Review

Orchid mycorrhiza: Isolation, culture, characterization and application

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A R T I C L E I N F O

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A B S T R A C T

In nature all orchids develop a mutualistic association with specialized group of fungi called orchid mycorrhiza. In natural environment, orchids are exclusively reliant on mycorrhizal fungi for seed germination, establishment, growth and development. In the early developmental stages, all orchids are mycoheterotrophic, and in most cases the association subsists throughout their lifespan. The significance of orchid mycorrhizal fungi (OMF) for seed germination, nourishment, growth promotion, and survival has been established more than a century ago and tremendous advancements were made in recent decades. Till date, understanding the mode of interaction between orchids and mycorrhizal fungi and determining the architecture of interaction networks are major challenges in ecology and evolutionary biology. The development of molecular techniques has empowered us to unscramble the fungal diversity comprised in the symbiosis and has facilitated us to overcome the difficulties associated with the conventional identification of OMF and orchid-fungal interactions. In addition, the application of modern biotechnological approaches resolved many controversial issues of OMF. Recently, OMF are used for biological hardening and growth promotion of *in vitro* raised seedlings, early flowering and quality improvement of flowers, and diseases and pests management. The aim of this communication is to update and summarize orchid mycorrhizal research including isolation, culture, and characterization of OMF, understanding the mechanisms of orchid-fungus symbiosis, and application of OMF in seed germination, growth promotion, and survival of asymbiotically raised seedlings and conservation of orchids. This appraisal will convey valuable insights to orchidologists engaged in biology, propagation, and conservation of orchids.

1. Introduction to mycorrhiza

Orchidaceae, the largest and most evolved plant family representing around 8–10% of flowering plants is more threatened than any other plant family. Orchids, the doyen amongst ornamentals, credited outstanding royalty as potted floricultural plant and cut flower. Development of new hybrids is comparatively easier in orchids because of their feeble reproductive barriers. In the meantime, more than 1,00,000 gorgeous hybrid orchids have been created and more than 1000 new hybrids are added every year (Hossain et al., 2013).

In natural environment a universal mutual interaction is developed between specific group of fungi and orchid roots called orchid mycorrhiza. Frank (1839–1900), a German botanist, first coined the term mycorrhiza (*mykorhiza*) meaning ‘fungus-root’ and described the phenomenon as ‘The entire body is neither tree root nor fungus alone, but like the thallus of lichens, a unique morphological organ which can be referred to as fungus-root (mycorrhiza)’ (Frank, 1885, translated by Yam and Arditti, 2009). Later on, he redefined the definition and redescribed the phenomenon as ‘Fungal hyphae grow organically together with roots forming a common organ which I named mycorrhiza’ (Frank, 2005). According to Pandey et al. (2013) mycorrhiza is a type of biotrophic, endophytic, mutualistic symbiosis prevalent in more than 90% of higher plants in which both associates benefited with regard to nutrients exchange, fitness and evolution. In course of time the research on mycorrhiza remarkably advanced and changed its definition such as, mycorrhizas are symbiotic relationships between higher plants and specific group of fungi dwell in the rhizosphere (Bellgard and Williams, 2011).

The mycorrhizal relationship is friendly, wide-ranging, and the fungi involved in the symbiosis are different in taxonomy, ecology and physiology. Mycorrhizal fungi have got acquaintance with almost all higher plants including crop plants, and therefore the association is ecologically, and economically very important. It is the most successful and important events of settlement by plants and the progression of biotrophic, root-inhabiting interdependences. More than 21,000 scientific papers and large number of books have been published on mycorrhizal associations since Frank coined the term “mycorrhiza” about 137 years ago (Trappe, 2005). Classically mycorrhizae were categorized into three major groups *i.e.* ectomycorrhiza (outside the root), endomycorrhiza (inside the root), and ectendomycorrhiza (both in- and outside the root). Based on the distinctive morphological features of root-fungus association there are six types of

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mycorrhizae have been described *viz*., (i) Arbuscular mycorrhiza (AM), (ii) Ectomycorrhiza (EM), (iii) Arbutoid mycorrhiza, (iv) Monotropoid mycorrhiza, (v) Ericoid mycorrhiza, and (vi) Orchid mycorrhiza (He et al., 2010). Seven types of mycorrhiza of equal taxonomic rank including ectendo-mycorrhizas with the above mycorrhizal diversity was described by Bellgard and Williams (2011).

Among the different mycorrhizal groups the orchid mycorrhiza is idiosyncratic type of mycorrhiza that exists in Orchidaceae only and it is economically, ecologically and evolutionarily very important. Based on mycorrhizal dependency orchids may be categorized into two groups as mixotrophic and myco-heterotropic/holomycotrophic. Most orchids are photosynthetic and therefore, their dependency on mycorrhizal fungi during seed germination stages only; such type of orchids is referred as mixotrophic orchid. The mixotrophic orchids could reduce their fungal dependency once they attained the photosynthetic activity. Although mycorrhizal associations prevail in all photosynthetic orchids throughout their life-cycle, the higher colonization was observed during their fast-vegetative growth and flowering time only. The achlorophyllous orchids entirely dependent on mycorrhizal fungi for nourishment throughout their life-cycle is called holomycotrophic. Although notable progresses have been made in orchid mycorrhizal research there is some crevice for indepth study of the orchid-fungal interaction framework. This review will provide opportunities for better understanding the diversity, ecological dynamics, isolation, culture, identification and application of OMF.

2. Brief history of orchid mycorrhiza

Early literature revealed that Link (1767−1851) might be the first botanist who perceived the occurrence of fungal endophytes in orchid roots by examining many tropical orchids including some European orchids. Before Frank's new term '*mykorhiza*' Link was working in Moscow and showed the incidence of fungi in orchid roots with a graphic illustration of a protocorm section of *Goodyera procera*, but he was unsuccessful to identify the fungus (Arditti, 1992). The practical importance of the fungi and their fundamental significance was first documented by Bernard (1899). The consequential research on orchid mycorrhiza was conducted by some renowned botanists like Schleiden von Reissek, Johann Georg Beer, Mathias Jacob Schleiden, Gaspard Adolphe Chatin, E̍douard, Ernest Prillieux, Hubert Leitgeb, Oscar Drude and others in different orchid species like *Corallorhiza, Epipogium, Goodyera, Limodorum, Angraecum maculatum, Cephalanthera grandiflora, Neottia* etc. and described some features of mycorrhizae (von Reissek, 1847; Beer, 1854; Schleiden, 1854; Prillieux, 1856; Leitgeb, 1864; Drude, 1873). Extensive studies on orchid mycorrhiza was conducted by Wahrlich (1886) who explained the peloton formation, digestion mechanism of the fungal pelotons and established the fact that mycorrhiza is universal in the Orchidaceae. Augustin, Dangeard and Armand critically investigated the mycorrhizal association of *Ophrys aranifera* and published two informative and interesting research paper. They provided excellent scientific illustrations and suggested that the mycobiont is a parasite but not harmful for host (Dangeard and Armand 1887, 1898). A critical investigation was conducted by MacDougal (1899) on orchid mycorrhizal association, especially that of *Corallorhiza* and *Aplectrum* and tried to draw conclusions on orchid fungal symbiosis. The incidence of fungal mycelia in the cortex cells of roots of *Corallorhiza innata, Neottia nidus-avis, Epipogon gmelini* and *Wullschlaegelia* was clearly visualized by Haberlandt (1914). However, none of the botanists listed above have drawn the precise conclusions regarding the role of the fungi associated with orchid roots. In support of this, it can therefore be said that they investigated orchid rhizomes and roots anatomy and observed only fungus like body in the cortex cell but they failed to identify fungal endophytes. The universal occurrence of OMF was resolutely established by Wahrlich (1886) through careful examinations of mycorrhizal roots from 500 different species orchids collected from diverse habitat of the world. Interestingly, the most significant and perceptive observation was accomplished by Bernard (1899) who described the role of fungi in orchids seed germination and stated that both fungus and host are benefited by each other. Nevertheless, the concept of plant-fungal interaction, ecology, physiology and evolution developed by Frank and Bernard is to date in the headway of recognition by scientific community (Trappe, 2005). Germinating seeds and seedlings of *Neottia nidus-avis* colonized by fungi was first observed by Bernard (1899). He also described that a fungal hyphal network is present in cortical cells and the epidermal cell layers are fungi free. He further noticed that fungal associations were common in all germinating seeds; the mycorrhizae were indispensable during the seed germination for *Neottia nidus-avis* and the plant was attached with the fungi in its all developmental stages. Initially he presumed that the fungi were parasitic to orchids but subsequently he provided more details that fungi can live without their host plants and the presence of fungi is required for orchids own development. Therefore, orchids are practically reliant on their fungal partner because they do not grow without the attendance of fungi. The second most exclusive works on orchid mycorrhiza was conducted by renowned German botanist Burgeff (1959) who was active with orchids for more than fifty years. He worked in a number of famous institutes such as the University of Halle, Bogor Botanical Gardens in Indonesia, Botanical Institute at the University of Wurzburg, University of Munich, University of Göttingen, etc. He mainly worked on the germination of terrestrial orchid's seed, orchid-mycorrhizal symbiosis and conservation. He developed techniques for symbiotic seed germination, isolation, culture and identification of fungal endophytes, and concluded that orchid root endophytes had distinct fungal group and named them Orcheomyces. He also proposed a self-conscious nomenclatural system such as *Mycelium radicis, Thrixspermum arachnites* following the name of the host orchid species. He also advocated the high specificity of orchid-fungal symbiosis because he was unsuccessful in asymbiotic germination of *Bletilla hyacinthine, Orchis*, and *Laelia* seeds. His perception is till date partially correct because many orchids require specific fungi for seed germination and many temperate orchids in general often do not germinate asymbiotically (Rasmussen, 1995). The logical conclusions of the Burgeff's failure might be the lack of providing appropriate culture medium and suitable growth conditions.

In natural environment, orchid roots may be colonized by various fungi, some of them may not be true mycorrhizal fungi. From the foundation work of mycorrhiza, *Rhizoctonia*-like fungi were thought to be the only mycobiont that develop orchid mycorrhiza. Later on, a number of fungal species were isolated and characterized from different orchids have shown similarity in many characteristics to *Rhizoctonia* spp., henceforth aggregately termed *Rhizoctonia*−like fungi (Warcup and Talbot, 1967; Moore, 1987; Ramsay et al., 1987; 1996; Andersen and Staplers, 1994; Shan et al., 2002; Sharon et al., 2008). Not only this, some saprophytic or plant pathogenic fungi, very much unrelated to orchid mycorrhiza such as *Penicillium, Trichoderma, Alternaria* etc. are also frequently found as root associated fungi in many tropical and temperate orchids (Fracchia et al., 2016). The *Rhizoctonia*-like fungi allied with orchids comprise free−living saprophytes (Burgeff, 1959) or opportunistic soil pathogens (Adams, 1988). In the past few decades, remarkable hand-outs on orchid mycorrhizal research have been published by some renowned orchidologists like, Curtis (1939), Hadley (1982), Warcup (1981), Warcup and Talbot (1967), Currah et al. (1987, 1997a,1997b), Clements (1988), Guo and Xu (1990a,1990b), Zelmer et al. (1996), Roberts (1999), Guo et al. (2000), Taylor et al. (2002), Selosse et al. (2004), Smith and Read (2008), Wu et al. (2009, 2010, 2011, 2013), Otero et al. (2002, 2007, 2013), Jacquemyn et al. (2011, 2014, 2015, 2017), Zhang et al. (1999, Zhang et al., 2012, 2020a, 2020b). In the present review, different aspects of orchid mycorrhiza were described giving emphasis on the isolation, identification, biology and application of OMF for propagation, growth promotion and conservation of orchids.

3. Basic biology of orchid mycorrhiza

In general, the biology of mycorrhiza means the function of diverse group of fungi with diverse group of plants, and broad range of habitats where the plants and fungi interact with each other. The association between plants and fungi are almost entirely mutualistic or symbiotic in which both partners enjoy benefits from each other. The reward seems typically related to enhanced nutrition of the host and protection of the infecting agents other than extend to chemical and physical protection, satisfy hormonal balance for host and modification of other rhizosphere microorganisms (Bellgard and Williams, 2011). In mycorrhiza, the association among the partner organisms is however remotely balanced. In general sense, it may be said that mycorrhizal relations are mutualistic, not parasitic, but words can not articulate straightforwardly the concept because there is a gradient from mutualism to parasitism. The mutual relationship regress and forwards along the mutualism-parasitism continuum when environmental conditions drastically change. Therefore, it is a fact that the benefits among the associates are not only quantifiably imbalanced but also qualitatively unequal; one partner enjoys extra benefits than the other. But it is not always clear which partner leans towards the parasitic or forced towards being parasitized. Frank (2005) hypothesized that mycorrhiza symbolizes a ubiquitous mutualistic symbiosis, in which the host and mycobionts nutritionally depend on each other; the fungi uptake hydro-mineral nutrition from surroundings and transport them back to the host; and the hosts, in return give protection and share photosynthesis products to fungi. Initially, this hypothesis was opposed by many of the scientific community. Although Frank unequivocally demonstrated the phenomenon, it takes long time to achieve conclusive validation (Trappe, 2005). Mycorrhizal endophytes colonized in the cortical cells of orchid roots and form condensed coils of mycelia, termed pelotons are supposed to be an adaptation mechanism in the host cells (Hadley, 1982) (Fig. 1A–C). Inside the host cells, the pelotons are bounded by a membrane (Peterson et al., 1998). The membrane is similar to the plasma membrane but lacks adenylate cyclase activity. During infection, the orientation of cell wall microfibrils and microtubules are altered that may be essential for modification of the cytoplasm and formation of membrane surrounding the pelotons (Peterson et al., 1998). Orchids are capable of controlling the fungal infection, and the mycobionts are adjusted to this control process. Upon mycorrhizal infection the activity of cellular enzymes for example peroxidase, ascorbic acid oxidase, polyphenol oxidase, and catalase are notably increased and digestion of fungal pelotons occurred (Blakeman et al., 1976). Orchids are able to regulate the extent of fungal infection and phytoalexins perform as mediators in this control process (Hadley, 1982). Orchinol, the first discovered phytoalexin in *Orchis* protocorms is considered to be inhibit fungal growth (Beyrle et al., 1995). It is now clear that the mature pelotons within the host cells digested due to the activity of lytic enzymes/phytoalexins/orchinol produced by the host and release nutrients from fungal body for host plants. Subsequently, new pelotons develop in the cells. The digestion and formation of pelotons are simultaneous processes (Fig. 1C). Some components of the digested pelotons act as fungal elicitors, which facilitate further colonization by fungi. There was evidence that fungal elicitors of orchid mycorrhiza are important in the growth and development of protocorms as well as the seedlings of orchids (Guo and Guo, 2001; Pan et al., 2004; Dong et al., 2008; Chen et al., 2008; Zhao and Liu, 2008). Fungal elicitors are a kind of bioactive substances delivered from fungal cell extracts or secretions, enriched with specific signal substances such as peptides, proteins, polysaccharides, chitins, fatty acids, glycolipids, glycoproteins, and other substances are high specifically and selectively induce specific plant genes expression to activate specific secondary metabolic pathways and accumulation of target specific secondary metabolites in the plant-fungi interactions (Kutney et al., 1993; Wu et al., 2006; Gong et al., 2018). These can increase biomass and secondary metabolites of plants and improve enzyme activity inside the plants. For example, *Fusarium oxysporum* induce the generation of taxol and fungal elicitors MF24 that promote the generation and accumulation of specific alkaloids (Gong et al., 2018). Elicitors also may possibly stimulate the protocorms to manufacture disease-resistance related compounds, such as L-phenylalanin ammonialyase (Hou and Guo, 2004; Luo and Yi, 2007). Some recent studies indicated that OMF could exuded several growth stimuli such as Indole-3-Acetic Acid (IAA), heteroauxin, dormin, zeatin, zeatin riboside, isoquercitrin, kinsenoside and gibberellins (GA_3), released into the extracellular space have pulsating effects in increasing the growth and development of orchids (Wu and Zheng, 1994; Wu et al., 2002; Yang et al., 2008; Shah et al., 2019; Ye et al., 2020). OMF metabolic products, even crude fungal extracts perform a significant role in germination of seeds, cell differentiation and protocorm development are also reported (Guo and Xu, 1990a, 1990b). However, not all fungal elicitors have promoting effect; some fungal elicitors have no promoting but even have inhibiting effects (Gong et al., 2018). Endophytic fungi evolve various strategies to overwhelm different obstacles during penetration to the hosts. Fungal hyphae can penetrate directly through epidermal cells of the root, develop intracellular hyphal network and establish a biotrophic collaboration with the host (Ye et al., 2020). Some recent reports documented that fungal endophytes can modify the secondary metabolites production in their host plants that are associated to increase enzyme activities, plant growth, and yield. For instance, Zhang et al. (2013) showed a noteworthy increase in shoot length, leaf numbers, dry weight of shoot, and isorhamnetin-3-*O*-β-D-rutinoside, kinsenosides, and isorhamnetin-3-*O*-β-D-glucopyranoside contents in *Anoectochilus formosanus* when co-cultured with *Mycena* sp. F-23. The endophytic fungus *Epulorhiza* sp. can increase the survival rate, fresh weight, dry weight, and activities of four enzymes (β–1,3-glucanase, chitinase, phenylalanine ammonialyase, and polyphenol oxidase) in the same orchid in open pot cultures (Tang and Guo, 2004).

4. Isolation of orchid mycorrhizal fungi

Mycorrhizal root selection, detection of fungal pelotons and colonized zones, size of root sections to be cultured, orientation of explants on the media surface and culture media are important issues for isolation of OMF. In epiphytic or lithophytic orchids, the root portions attached with the substratum usually showed the greater occurrence of mycorrhizal associations but in terrestrial orchids, comparatively old roots showed the greater occurrence. The mycorrhizal zones of the roots are usually yellowish or brownish in color and aerial portions and root tips are devoid of mycorrhizal occurrence (Arditti, 1992; Porras–Alfaro and Bayman, 2003; Hossain, 2019). Root sections used for isolation of fungal endophytes by different researchers were of varied thickness (0.5 mm to 2 cm). Now-a-days, fungal colonized cells from the root cortex, a tuft of pelotons or single peloton are used as a promising source for isolation of fungal endophytes (Kristiansen et al., 2001a; Wright et al., 2010; Chen et al., 2011; Long et al., 2014; Khamchatra et al., 2016). The single peloton suspensions are prepared by abrading mycorrhizal root into sterile deionized water and then spread over the medium and incubated for 4–8 d at 21 °C for fungal growth. Hyphal bits from fast-growing zones are sub-cultured successively onto fresh culture plates to get pure culture.

For growth of fungal endophytes, different culture media were used by different researchers such as Potato Dextrose Agar (PDA), Cornmeal Agar, Oatmeal Agar, Malt Extract Agar, Bran Agar, Water

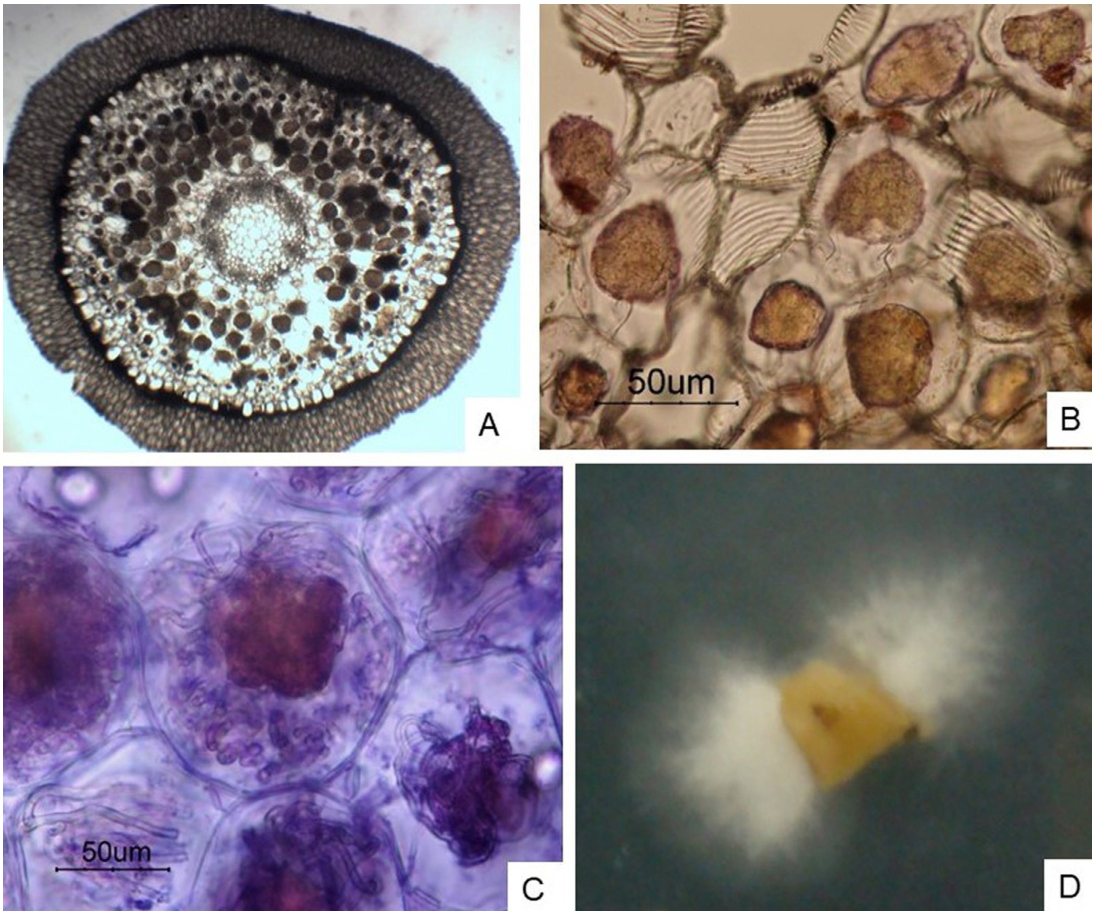

Fig. 1. Fungal pelotons in orchid root sections: (A) *Geodorum densiflorum*, (B) *Rhynchostylis retusa*, (C) *Phaius tenkervillae* and (E) Fungi growing from orchid root cortex on PDA medium. (unpublished photographs of author).

Agar, and Molten Fungal Isolation Medium (FIM) but PDA was the most common. Some broad-spectrum antibiotics for instance streptomycin, tetracycline, penicillin and chloramphenicol were exercised in the fungal growth medium at various concentrations (20–150 μg/ml) to control the growth of bacteria, attached to the surface of the root sections or associated with mycorrhizal endophytes (Otero et al., 2002; Zhu et al., 2008; Wright et al., 2010; Hossain et al., 2013). Different researchers applied different techniques for isolation of fungal endophytes. Clumps of cells with fungal pelotons were used by Currah et al. (1987) to isolate fungal endophytes. *Rhizoctonia*-like fungi were isolated from peloton suspension culture in molten agar medium fortified with 50 μg/ml streptomycin and 20 μg/ml tetracycline (Warcup and Talbot, 1967).

A number of fungi were characterized from *Eulophia flava, Habenaria dentata, Goodyera procera*, and *Spiranthes hongkongensis* through culturing individual root sections on PDA containing 50 mg/L streptomycin and 3 mg/L rose bengal by incubating in dark condition at 25 °C (Shan et al., 2002). Velamen tissues of *Vanilla planifolia* were used by Porras–Alfaro and Bayman (2007) for isolation of OMF on PDA medium enriched with 50 μg/ml each of penicillin and streptomycin. Similar method was also followed by Otero et al. (2002) for isolation of fungal endophytes from *Campylocentrum filiforme, C. fasciola, Erythrodes plantaginea, Ionopsis utricularioides, I. satyrioides, Oncidium altissimum, Oeceoclades maculata, Psychilis monensis* and *Tolumnia variegata*. A little different method was exercised by Zhu et al. (2008) in *Cremastra appendiculata, Pleione yunnanensis* and *P. bulbocodioides*. Root hairs, epidermis and velamen tissues were removed from roots and mycorrhizal portion with pelotons were screened by microscopic inspection; disinfected by dipping in 150 μg/ml potassium Penicillin G and 150 μg/ml streptomycin sulfate solution, cut into small segments and grown on PDA. More than two antibiotics *i.e*. 50 μg/ml streptomycin, 50 μg/ml oxytetracycline, and 50 μg/ml penicillin was also used in PDA medium for isolation of fungal endophytes from *Cymbidium, Dendrobium*, and *Paphiopedilum*, (Nontachaiyapoom et al., 2010). Streptomycin sulfate at 0.75 μg/ml concentration in PDA medium was proved effective for isolation of *Ceratobacidium* AM and *Ceratobasidium* RR respectively from fresh roots of naturally grown *Aerides multiflorum* and *Rhynchostylis retusa* (Hossain et al., 2013) (Fig. 1D). A different procedure was described by Fracchia et al. (2016) for isolation of OMF from a rare terrestrial orchid, *Chloraea riojana*. In this procedure glass slides containing orchid seeds were packed with double mesh (90 μm) and buried in pots containing collected rhizospheric soil and checked periodically under a binocular microscope to observe protocorm development. The protocorms infected with fungi were rinsed with sterile water and transferred to culture plates with PDA fortified with antibiotics to allow fungal growth. *In situ* seed bating technique was applied for isolation of endophytic as well as OMF from seeds of *Dendrobium friedericksianum* (Khamchatra et al., 2016). Approximately 300–400 seeds were taken in a small pouch of nylon mesh (5 cm × 10 cm) with 45 μm pore. The pouches were embedded singly on the center of 35 mm plastic sheet and placed onto tree bark close to mature roots of *D. friedericksianum*. These were then coated with a layer of sterile moisten *Sphagnum* to avert desiccation and protected by gutter mesh on tree trunk. Occasional inspections were made to observe the germination of seeds and fungal endophytes were isolated from seed originated protocorms. The surface of seed packets was sterilized by washing under running tap water and dipping in 70% ethanol for 15–30 s. The infected protocorms were then detected with the help of a dissecting microscope and disinfected by dipping them in 0.5% NaOCl solution for 3–5 min and rinsing thoroughly with sterilized distilled water. Finally, protocorms were cultured on the surface of Czapek Dox agar medium augmented with 0.05 g/L streptomycin. Pure cultures were raised by successive subculture of the fungi that grown from the cultured protocorms onto fresh PDA medium.

5. Characterization of orchid mycorrhizal fungi

Thousands of Basidiomycota and some Ascomycota fungi form association with tropical, temperate and Mediterranean orchids (Suz et al., 2018). Approximately 100 *Rhizoctonia*-like OMF including some new species such as *Mycena anoectochila, M. dendrobii, M. orchidicola* have been isolated, identified and validated their role in orchid propagation and conservation. In addition, some non-mycorrhizal fungi such as *Trichoderma* spp., *Aspergillus* spp., *Fusarium* spp., *Piriformospora indica, P. Williamsii, Serendipita vermifera, Gymnopus* sp. etc. have also been isolated from orchids; most of them belong to Basidiomycotina and Deuteromycotina (He et al., 2010; Fracchia et al., 2016; Weiß et al., 2016; Thakur et al., 2018; Li et al., 2022). An extensive study was made on 44 orchid species in China that represented diverse life forms and co-occurrence in tropical forests (Xing et al., 2019). A total of 1343 diverging sequences were generated through ITS regions sequencing from typical orchid mycorrhizal fungi. Among the sequences, 1324 matched to Basidiomycetes and only 19 sequences belonged to Ascomycetes fungi. The sequences of Basidiomycetes fungi yielded a total of 87 different fungal operational taxonomic units (OTUs), the majority of which were Tulasnellaceae and a few were Ceratobasidiaceae and Sebacinales. The other fungal species retrieved from orchids were members of the Cortinariaceae, Marasmiaceae, Russulaceae, Thelephoraceae, and unknown Cantharellales and Atractiellales. Furthermore, a number of conceivably endophytic fungi fitting to Septobasidiaceae and Tricholomataceae were also infrequently detected.

The characteristics of orchid mycorrhizal fungi has been studied through morphology by using optical, scanning and electron microscopy (Athipunyakom et al., 2004; Jakucs et al., 2005; Zhu et al., 2008; Suryantini et al., 2015; Jiang et al., 2019; Hossain, 2019), and molecular analysis of genomic DNA (Taylor and McCormick, 2008; Fracchia et al., 2016; Thakur et al., 2018; Xing et al., 2019; Zhang et al., 2020a, 2020b; Li et al., 2022). The fungal partner in orchid mycorrhiza generally belongs to Basidiomycetes and the isolates usually produce sterile mycelia and exhibit very little sporulating tendency in artificial culture condition. Colony growth, color of young and mature colony, affinities for hyphal anastomosis and production of monilioid cells are distinctive morphological features for identification of OMF. For typical *Rhizoctonia*-like OMF, the young colonies are generally cottony white or creamy white that turned to light brown to deep brown at maturity (Fig. 2A). Certain microscopic features, particularly the size and shape of monilioid cells (Fig. 2B), formation of sclerotia, the ultrastructure of the septal pore (dolipore septum), right angle branching, a constriction near branching point, a dolipore septum at the base of branch, cytomorphology of the vegetative and reproductive hyphae, number of nuclei in both vegetative and monilioid cells, teleomorphs, and enzyme activity are useful identifying characters for sterile mycelia of *Rhizoctonia* complex (Shan et al., 2002; Hossain et al., 2013; Jiang et al., 2015; Hossain, 2019). Interestingly, most of the orchid mycobionts encompass the above characters. However, the mycelia generally do not produce many unique characteristics that enable them to identify below the generic level which is now possible by means of nrDNA analysis of the orchid endophytes.

As already mentioned that only *Rhizoctonia*-like fungi develop orchid mycorrhiza but recent investigations have publicized that some other fungal groups can also be associated with orchids. For instance, members of the Basidiomycetes ‘rust’ lineage are mycobionts in some orchid species (Kottke et al., 2010). Ascomycetes fungi related to the ectomycorrhizal Septomycetes along with truffles were reported to associate with *Epipactis microphylla* (Selosse et al., 2004). Along with *Tulasnella violea* and *Epulorhiza repens; Trichosporiella mutisporum* and *Fusarium* were associated as root endophytes in *Dendrobium friedericksianum* (Khamchatra et al., 2016b). Many *Rhizoctonia*-like OMF has two phases in their lifetime *i.e.* anamorphic and teleomorphic phases. Anamorphs are mycelial sterilia, while teleomorphs produced distinctive reproductive features. Fifteen *Rhizoctonia* anamorphs were listed belonging to Heterobasidiomycetes are known to be orchid symbionts: 5 species of *Ceratobasidium*, 3 species of *Thanatephorus*, 5 species of *Tulasnella*, 1 species of *Serendipta* (*Sebacina*) and *Oliveonia* (Roberts, 1999). Another report stated 21 endophytic fungi characterized from the roots of *Goodyera procera, Eulophia flava, Spiranthes hongkongensis* and *Habenaria dentata* were *Rhizoctonia* complex (Shan et al., 2002).

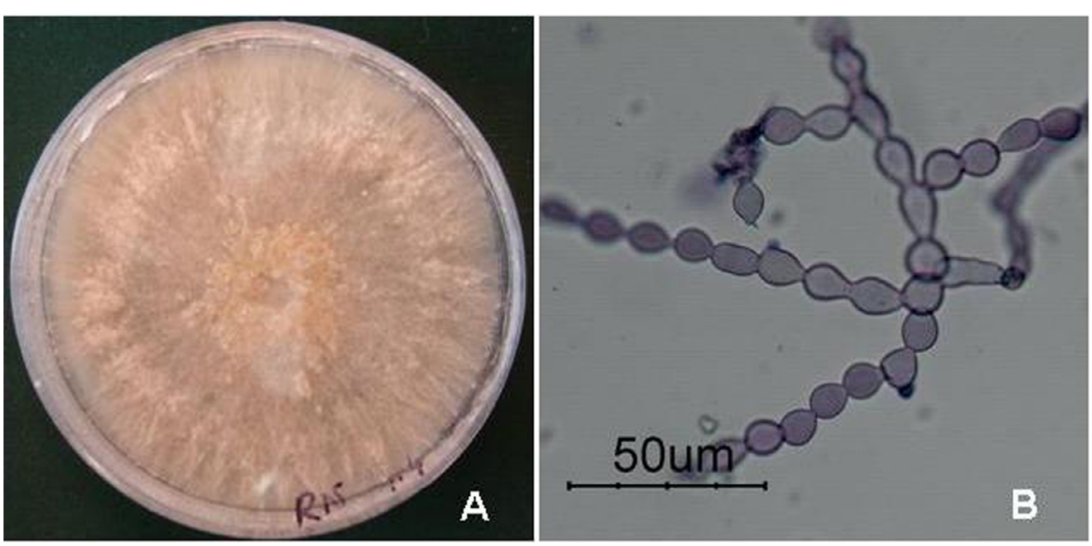

Fig. 2. Characteristics of OMF: (A) Mature colony (21 days old) of mycorrhizal fungi isolated from *Rhynchostylis retusa*, and (B) Monilioid cells of OMF after stained with lactophenol-cotton-blue. (unpublished photographs of author).

For studying microscopic features of OMF various staining procedures were applied for example, trypan blue or aniline blue, lactophenol triglycero-cotton blue, 4′, 6′ diamino-2-phenylindole (DAPI), orcein, HCl-geimsa and acridine orange etc. The size and shape of monilioid cells and hyphal diameter are usually measured by staining with lactophenol triglycero-cotton blue / safranin 0.5% (w/v) and 3% KOH (1:1) / trypan blue. Growth rate of fungal colony are measured by inoculating a bit of live mycelia at the center of petri plates and determine radial increases at certain time periods (Shan et al., 2002; Nontachaiyapoom et al., 2010; Hossain, 2019; Bhuiyan et al., 2021). Nuclei number in hyphal as well as monilioid cells are determined by fixing a bit of fungal mat in 2% formaldehyde solution for 2 min, cleaned for 1 min by deionized distilled water, and subsequently staining with Gold-antifade DAPI or singly DAPI stain for 10 min and de-stained by washing for 2 min with deionized distilled water and observed by microscope with UV light keeping in 50% glycerol solution (Shan et al., 2002; Hossain, 2019).

The *Rhizoctonia*-like fungi may also be characterized by investigating right-angle branching pattern, a septum near the branching point, a constriction at the branching point, and ultra-structure of dolipore septum. Number of nucleus in vegetative and monilioid cells is another significant morphological character for identification of *Rhizoctonia*-like orchid endophytes (Shan et al., 2002; Hossain et al., 2013). Multi-, bi- and uninucleate cells have been reported in orchid mycorrhizal endophytes. *Rhizoctonia solani* (teleomorph: *Thanatephorus*) and *Ceratobasidium* sp. strain GC possesses multinucleate cells (Hossain, 2019). *Ceratobasidium* and *Tulasnella* which comprise the most common group of OMF have bi-nucleate cells (Currah et al., 1997a, 1997b; Hossain et al., 2013; Jiang et al., 2015). Uni-nucleate strains were also reported in *Ceratobasidium* spp. but rarely found in orchids (Hietala et al., 2001). Based on the nuclear number and septum ultra-structural, parenthesome perforation, and teleomorphic stage, *Rhizoctonia* and *Rhizoctonia*-like fungi assigned as three new genera *i.e. Monilioipsis Epulorhiza* and *Ceratorhiza* (Moore 1987; Ma et al., 2003; Garcia et al., 2006). This framework was adopted by several researchers for the identification of OMF that ratifies a taxonomically acceptable and justified method to designate taxa (Shan et al., 2002). Thus, species concept can be confirmed within these genera based on the specific cultural features *e.g.* mycelial morphology and affinities for hyphal anastomosis (mating of hyphae) on specific media (Currah and Zelmer, 1992; Vilgalys and Cubeta, 1994; Zelmer and Currah, 1995; Andersen, 1996; Jiang et al., 2015).

The genus *Rhizoctonia* represented taxonomically diverse groups that differ in anamorphic and teleomorphic stages (Warcup and Talbot, 1967; Currah et al., 1987). However, they belong to three different basidiomycetes fungi families, *i.e.*, Ceratobasidiaceae, Tulasnellaceae and Serendipitaceae (Jacquemyn et al., 2017; Li et al., 2022). The genus *Rhizoctonia* was first proposed by De Candolle (1815) which included a group of heterogeneous filamentous fungi that generally do not produce sexual spores and share only limited general characteristics in their anamorphic phase (Garcia et al., 2006). Now it is clear that many of these fungi are capable to produce both asexual and sexual state. The asexual phase of these fungi is termed anamorph, while the sexual phase is called teleomorph and the entire fungus with its forms and features encompasses the holomorph (both teleomorph and anamorph). Some of these fungi have no known teleomorph or anamorph. Interestingly, several fungi, especially some Basidiomycota and Ascomycota develop anamorphic and teleomorphic features at certain stages of their life cycle. That's why, two different names were given for several fungi, one for the anamorph and the other for the teleomorph. The holomorph is more accurately referred to by their teleomorphic names when the anamorphs are allied to their corresponding teleomorphs. For instance, the teleomorph or sexual state of *Tulasnella* spp. has asexual state which called *Epulorhiza* spp. (Syn. *Rhizoctonia* spp.) is very common as orchid mycorrhizal fungi. Similarly, *Thanatephorus praticola, T. cucumeris*, and *Waitea circinata* are teleomorphs of *Rhizoctonia praticola, R. solani*, and *R. circinata* respectively. The holomorph of these fungi are correctly referred to as *Rhizoctonia* spp. Therefore, the *Rhizoctonia* complex comprises the anamorphic genera *Epulorhiza, Ceratorhiza, Moniliopsis*, and teleomorphic genera *Ceratobasidium, Thanatephorus, Tulasnella*, and *Sebacina* (Warcup and Talbot, 1967) which are universally identified as orchid mycorrhizal fungi. Development of chains of swollen, inflated, globular/spherical asexual resistant propagules called monilioid cells is another significant anamorphic feature of *Rhizoctonia*-like OMF (Shan et al., 2002, Hossain et al., 2013).

Another way of classifying fungi is by studying anastomosis groups (AG). Vegetative hyphae of two different fungal species of the same AG have the ability to fuse together. This traditional hyphal mating system was widely employed to identify and classify *Rhizoctonia*-like fungi. It is still a valid identification method which genetically supported by modern DNA-based biomolecular techniques (Sharon et al., 2008; Jiang et al., 2015). Hyphal anastomosis affinities *i.e*. rejoining of hyphae (AGs) have been studied by different authors in *Rhizoctonia solani*. The bi-nucleate *Rhizoctonia* fungi included 21 AGs while the uni-nucleate *Rhizoctonia* fungi included a single AG (Sneh et al., 1991; Hietala et al., 1994; Sen et al., 1999). Two anastomosis groups *i.e*. AG-6 & AG-12 belong to the multinucleate *Rhizoctonia* spp. are known to be associated with orchids (Carling, 1996; Carling et al., 1999; Pope and Carter, 2001). Nonetheless, the most common OMF are bi-nucleate *Rhizoctonia* spp. (Currah et al., 1997a, 1997b; Shan et al., 2002) and uni-nucleate *Rhizoctonia* has rarely been reported from orchids. The vegetative hyphal fusion behavior (AG), intra-specific group (ISG) among AGs, morphology, pathogenicity, etc. were adopted as the most common techniques for identification of *Rhizoctonia* spp. and examined their genetic variation (Vilgalys and Cubeta, 1994; Shan et al., 2002; González-Chávez et al., 2018).

The conventional methods for identification of OMF have some confines since the fungi coupled to orchids are produced sterile mycelia. Accordingly, the wide vegetative features for the documentation of fungi may result in paraphyletic taxonomy because numerous unrelated fungi are breathing together. For example, anastomosis reactions are in fact insufficient for accurate identification and classification of some *Rhizoctonia* isolates. There are some difficulties to determine the appropriate AGs also. For instance, some fungi have lost their copulating ability even to self-anastomose (Hyakumachi and Ui, 1987). Some multinucleate *Rhizoctonia* isolates of the AG2 had no anastomosis response with certain isolates of the same AG (Carling et al., 2002). Similarly, some bi-nucleate *Rhizoctonia* isolates are unable to develop anastomosis with definite representative fungi of the similar AG (Sneh et al., 1991; Martin, 2000). Conversely, some binucleate or multinucleate *Rhizoctonia* isolates form anastomose with more than one representative isolates (Ogoshi et al., 1983; Sneh et al., 1991; Carling, 1996; Carling et al., 2002). The above limitations of morphological identification necessitate the practice of advanced bio-molecular methods for accurate recognition (Sen et al., 1999; Shan et al., 2002; Otero et al., 2002; Yagame et al., 2008; Thakur et al., 218). Amplification of specific DNA segments from a single fungal peloton through direct PCR using specific primer sets were exercised for the identification of fungi which provided comprehensive and thoughtful insights of the symbiotic relationship between orchids and OMF. The random amplified polymorphic DNA (RAPD) and restriction fragment length polymorphism (RFLP) are also customarily applied for description of OMF (Sharon et al., 2008; Wright et al., 2010). The RAPD analysis includes the usage of single short primers of random nucleotide sequences to repetitively amplify targeted genomic DNA fragments. These short primers designated as genetic markers are exercised to detect polymorphisms amongst the amplified nucleotide sequences which are stained with ethidium bromide dye and visualized as bands after electrophoresis through an agarose gel. Theoretically, RAPD was illustrated as a simple form of PCR using random primers to distinguish DNA polymorphism and for expediency (Bairu et al., 2011). The AFLP is also a PCR-based technique frequently exploited in genetics, DNA fingerprinting, and genetic engineering programs. It was established at the beginning of the 1990s to overwhelm the constraints of reproducibility related to RAPD. Hypothetically, AFLP is an ingenious combination of RFLP and the flexibility of PCR-based knowledge (Agarwal et al., 2008). It offers a very influential and novel DNA fingerprinting method in genomic research for any origin and complexity of DNAs.

Sequencing and analysis of ITS regions of nuclear ribosomal DNA is the widely used and more powerful molecular technique for accurate identification of OMF (Table 1) (Sharon et al., 2008; Steinfort et al., 2010; Wu et al., 2010; Zhang et al., 2012; Jacquemyn et al., 2011; Thakur et al., 218). This technique is carried out by isolating nrDNA of fungal isolates, amplify the ITS regions by PCR with specific primer (for example ITS1, ITS2, ITS3, ITS4, ITS5 etc.), and then compare the sequences with the accessible information from Gene Bank (Fig. 3). The ITS regions have some important characteristic features which brand it as a universal ‘barcode’ for documentation of the broadest range of fungi, such as (i) due to the high copy number ITS regions are practically easy to amplify, (ii) comparatively a few primers are required to amplify the highly conserved large subunit (LSU) and small subunit (SSU) flanking regions, (iii) differs slightly within the species but vastly between the species, and (iv) well-represented DNA sequences in the Gene Bank data than any other genes of the fungal genome (Schoch et al., 2012; Sun and Guo, 2012). Although ITS region sequencing and analysis is widely used to identify fungi for over a decade, it has a few limitations. According to DeSalle et al. (2008), a single barcode is inadequate for the representation of the panorama of a fungal population, and amalgamations of several barcodes were thus recommended. Recently, by applying next generation sequencing technologies (NGS) with specific DNA barcoding primers and metagenomics analysis successfully identified accurate fungal strains and understand the biological interactions between orchids and fungi (Huang et al., 2014; Yeh et al., 2019). The major strength of NGS is its higher sensitivity to detect low-frequency variants and the method can detect abnormalities across the entire genome using less DNA than required for traditional DNA sequencing approaches. Moreover, the rapid advancement of NGS and functional genomics methodologies contribute to identify genes engaged in the mycorrhizal association and unscrambling the molecular strategies regulating this archetypal symbiosis (Yeh et al., 2019).

6. Role of OMF in seed germination

The orchid seed are the smallest seed in nature which have no organized embryo and sufficient food reserve for germination. Therefore, a symbiotic association with compatible mycorrizal fungi is obligatory for natural seed germination. It is believed that the fungi provide hydro-nutrients from surrounding environment, such as carbon resources, nitrogen, phosphorus, vitamins and growth stimulus that meet up the nutrient demand of germinating seeds (Rasmussen, 1995; Smith and Read, 2008). Some experimental reports support this thought, for instance, Yeh et al. (2019) reported that fungal exudates contain stimulus substances that stimulate seed germination and enhance germination percentage *in vitro*. The symbiotic germination technique was successfully applied in terrestrial orchids with specific fungi as an alternative method. It was also observed that *Orchis* and *Dactylorhiza* species are better germinate only in the attendance of fungi (Clements, 1988). Symbiotic germination of seed has been proved advantageous in propagation of many European, Australian, and North American terrestrial orchids. However, it was not always proficient and fruitful in all tested orchids (Zettler, 1997). Research on symbiotic seed germination proclaim that the orchid-fungal interaction was not highly species-specific, although Arditti (1992) suggested that it might be true for some orchid species. Symbiotic seed germination experiments among 15 European and North American orchids species revealed that some orchids responded to diverse fungi and some fungi persuaded germination in more than one orchid species (Smreciu and Currah, 1989). Interestingly, the endangered terrestrial orchid *Isotria medeoloides* was unsuccessful in symbiotic germination with the fungi that naturally occurred with this orchid.

Seed germination, ecological fitness, dependency on external nutrition supply, growth and development of many orchids *viz, Arundina chinensis, Anoectochilus roxburghii, Caladenia* spp., *Cephalanthera falcata, Changnienia amoena, Corallorhiza odontorhiza, Cymbidium ensifolium*, C. goeringii, *C. lianpan, C. mastersii, C. sinense, C. tracyanum, Cremastra appendiculate, Cyrtopodium glutiniferum, Dactylorrhiza* spp., *Dendrobium brymerianum, D. chrysotoxum, D. candidum, D. catenatum, D. densiflorum, D. hancockii, D. loddigesii, D. nobile, D. officinale, D. moschatum, D. primulinum, Epigeneium rotundatum, Epipogium roseum, Encyclia tampensis, Erythrorchis ochobiensis, Gastrodia elata, G. confusoides, G. nantoensis, G. appendiculata, Goodyera pubescens, G. repens, Habenaria macroceratitis, Liparis liliifolia*, L. *viridiflora, Orchis anthropophora, O. militaris, O. purpurea, Platanthera bifolia, P. clavellata, P. praeclara, P. integrilabia, Paphiopedilum armeniacum, Pleione bulbolodiodes, Spathoglottis plicata, Spiranthes hongkongensis, Taeniophyllum obtusum, Tolumnia variegata, Vanda amesiana* etc. were enhanced by treating with different OMF such as *Ceratobasidium* spp., *Epulorhiza* spp., *Ceratorhiza* spp., *Rhizoctonia* spp., *Tulasnella* spp. *Mycena anoectochila, M. dendrobii, M. osmundicola*, and *M. orchidicola*. Seed germination and seedling development were strongly stimulated in some orchid species by *Ceratorhiza, Epulorhiza* and some *Rhizoctonia* isolates (Smreciu and Currah, 1989; Chang, 2006, 2007, 2008; Jiang et al., 2015). Although fundamental roles of orchid root associated fungi on germination of seeds and sprout development in orchids was discovered in the early 20th century and recent decades rigorous studies on different aspects of OMF has been carried out, some aspects of orchid mycorrhizal relationships have not yet been elucidated (Kulikov and Filippov, 2001; Dearnaley, 2007; Yeh et al., 2019). Experiments on symbiotic seed germination were carried out in *Tolumnia variegata* which was linked with a relatively broad phylogenetic range of fungi, including four clades of *Ceratobasidium* spp. and some un-identified fungi (Otero et al., 2005). Significant difference was observed for both symbiotic germination and seedling development among *T. variegata* populations. Significant seed germination and seedling development was induced by two clades *Ceratobasidium* fungi. Combining luminous reasoning with skillful experimentation they concluded that geographic distribution of mycorrhizal fungi is not homogenous which may affect orchid distribution, population dynamics and evolutionary ecology. Although a number of fungi associated with orchids, only a few show species specificity (Dearnaley, 2007).

7. Species-specificity of orchid mycorrhiza

Species−specificity of OMF has been a subject of interest for long time (Curtis, 1939; Clements, 1988; McKendrick et al., 2000, 2002; Taylor et al., 2002; Otero et al., 2002, 2007; Kennedy et al., 2011). Several earlier workers believed that the orchid-fungal relationship is specific while others advocated that it is not factual for all orchids (Arditti, 1992). Most of the investigations on orchid−fungus specificity was carried out *in vitro* as a part of symbiotic seed germination tests; only a few experiments were carried out on germination in natural conditions. Many researchers suggested that it is not possible to draw a conclusion from only *in vitro* experiments without testing field performance. Moreover, isolation and characterization of mycorrhizal fungi from different orchid taxa and cross infection

Table 1
Some published records of orchid fungal endophytes and methods of identification.

Orchid species	Associated Fungi	Method of identification	Refs.
Dendrobium officinale	Five *Tulasnella* species and one *Fusarium* species	The fungal strains were identified morphologically and then grouped based on culture characteristics such as colony morphology, hyphal characteristics, hyphal color, and the colours associated with secondary metabolites. Molecular identification was done through PCR amplification and sequencing of ITS regions of rDNA and the 5.8S gene.	Chen et al., 2021
Dendrobium officinale	Isolated nine OMF, among them seven fungi belong to Tulasnellaceae *i.e. Tulasnella calospora* JM, *Tulasnella calospora* TG1, *Tulasnella calospora* TG3, *Tulasnella calospora* TY3, *Tulasnella* sp. TY4, Tulasnella sp. TG2 and, Tulasnellaceae TY2, and the other two fungal strains were Sebacinaceae TY1 and Sebacinales LQ.	The fungal strains were identified morphologically by observing and recording the culture characteristics on PDA medium such as color and surface shapes. The fungal strains morphologically looked like OMFs were molecularly identified through sequencing and analysis of the rDNA region containing the two ITS regions and the 5.8S gene.	Wang et al., 2021
Paphiopedilum spicerianum	Two *Tulasnella* spp.	Identified morphologically by observing white colonies, presence of monilioid cells and rough with branched and septate hyphae. Molecular identification was done through sequencing and analysis of ITS regions of rDNA.	Yang et al., 2021
Rhynchostylis retusa	*Ceratobasidium ramicola*	Identified morphologically by studying cultural and microscopic features, such as colony appearance, colony color, presence of monilioid cells, diameter of vegetative hyphae, right angle branching pattern of the fungal endophyte etc. Molecular identification was done through sequencing and analysis of ITS regions of rDNA.	Bhuiyan et al., 2021
Anoectochilus roxburghii	Isolated an OMF *Ceratobasidium* sp. AR2	OMF was identified based on morphological features and phylogenetic tree analysis of the ITS sequences.	Zhang et al., 2020a
Cypripedium candidum, Platanthera lacera, P. leucophaea, P. paramoena, Spiranthes magnicamporum, and *S. vernalis*	Isolated 39 OMF in which 31 (79%) were assignable to the genus *Ceratobasidium*, and other 8 species were *Tulasnella*.	Molecular identification involving ribosomal DNA internal transcribed spacer (ITS) amplification and Sanger sequencing.	Thixton et al., 2020
Dendrobium officinale	Isolated six OMF *Sebacina* strains S2 and S3 and *Tulasnella* strains S4, S5, S6 and S7	Mycobionts were identified based on morphological features and phylogenetic tree analysis of the ITS sequences.	Zhang et al., 2020b
Pleurothallis coriacardia	Fungal isolates identified as *Ilyonectria, Coprinellus, Nigrospora, Chaetomium, Fusarium, Trametes, Trichoderma*, and Unidentified Endophyte.	Fungal endophytes were identified through ITS sequence analysis of rDNA.	Maldonado et al., 2020
Epidendrum fruticosum, E. geminiflorum, Epidendrum sp.1, *Epidendrum* sp.2, *Epidendrum* sp.3, *Lepanthes* sp., *Frondaria caulescens, Pleurothallis coriacardia, Odontoglossum* sp. and *Stelis* sp.	*Coprinellus radians, Trametes* sp., *Meyerozyma guilliermondii, Penicillium chrysogenum, Penicillium rubens, Fusarium* sp., *Botryobasidium* sp. and Lepidiotaceae	Identified through sequences analysis of the ITS of nuclear rDNA	Salazar et al., 2020
Dendrobium exile	Seven different fungal species including three *Rhizoctonia* fungi (*Tulasnella* sp. DesI, *Tulasnella* sp. DerIV and *Tulasnella* sp. DerV), three non-*Rhizoctonia* fungi (*Nodulisporium* sp. DerI, *Xylaria plebeja* DerII, *Colletotrichum* sp. DerIII) and one unidentified fungus DesII were isolated and identified.	The fungal strains were identified by both morphological (culture characteristics on PDA medium such as color, surface shape, tissue texture and microscope features) and molecular techniques (ITS sequencing).	Meng et al., 2019a
Dendrobium moniliforme	*Tulasnella* spp. and *Rigidoporus* sp.	All fungal strains were characterized molecularly by internal transcribed spacer (ITS) region sequences of nuclear ribosomal DNA using the ITS1 and ITS4 primers.	Meng et al., 2019b
Forty-four orchid species belong to *Acampe, Aerides, Bulbophyllum, Cleisostoma, Coelogyne, Eria, Eulophia, Flickingaria, Goodyera, Hebanaria, Hetaeria, Lusia, Liparis, Gastrochilus, Malaxis, Nervilia, Oberonia, Epigeneium, Panisea, Pholidota, Pelatantheria, Rhomboda,*	Mainly Tulasnellaceae, a few Ceratobasidiaceae and Sebacinales. Some other fungal taxa belong to Thelephoraceae, Cortinariaceae, Marasmiaceae, Russulaceae, Tricholomataceae and Septobasidiaceae were also retrieved from	Molecular identification through amplification and sequencing of internal transcribed spacer (ITS) of ribosomal DNA and phylogenetic diversity analysis of fungal associations between orchid and fungi.	Xing et al., 2019

(*continued*)

Table 1 (*Continued*)

Orchid species	Associated Fungi	Method of identification	Refs.
Rhyncostylis, Sarcoglyphis, Sunipia, Thelasis, Xeuxine etc.	some these orchids.		
Gastrochilus calceolaris	Isolated and identified a multinucleate *Rhizoctonia*-like OMF *Ceratobasidium* sp. GC.	Both morphological (hyphal diameter, nuclear number, branching pattern, dolipore septum etc.) and molecular approaches (ITS sequencing of nrDNA) were applied for proper identification.	Hossain, 2019
A. praemorsa	five endophytic fungal species, *Trichoderma asperellum, Trichoderma atroviride, Trichoderma harzianum, Diaporthe eucalyptorum* and *Endomelanconiopsis endophytica* isolated from the velamen roots	The endophytic fungi were identified by morphological and molecular methods.	
Crepidium acuminatum (Syn. *Malaxis acuminata*)	Isolated an orchid mycorrhizal symbiont *Tulasnella* and two associated fungi, *Aspergillus* and *Penicillium.*	Both morphological (Scanning and electron microscopy) as well as molecular approaches *i.e.* sequences analysis of the ITS of nuclear rDNA. using a variety of primer combinations.	Thakur et al., 2018
Cyrtochilum retusum and *Epidendrum macrum*	83 OMF operational taxonomic units (OTUs) belonging to Tulasnellaceae, Ceratobasidiaceae, Serendipitaceae and Atractiellales were described.	The OMF community composition was determined using the Illumina MiSeq sequencing of the internal transcribed spacer 2 (ITS2) regions.	Cevallos et al., 2018
Codonorchis lessonii	Three different fungal strains were isolated and identified belonging to Ceratobasidiaceae and Tulasnellaceae associated with this orchid.	Morphological (cultural characters, hyphal characters *i.e.* strait angle ramification, simple septa, constriction at the base of a branching hypha, and presence of monilioid cells) and molecular characterization through sequencing of ITS region of nuclear rDNA and phylogenetic analysis.	Pereira et al., 2018
Bletia roezlii, B. purpurata, and *B. punctata*	Isolated 39 fungal strains of '*Rhizoctonia*-like fungal complex' all belong to four subgroups of the genus *Tulasnella* J. Schröt	Morphological characters and molecular phylogenetic analysis.	Nambo et al., 2018
Dactylorhiza romana subsp. *georgica, D. umbrosa, Orchis pinetorum, O. spitzelii, O. coriophora, O. collina, O. anatolica, O. simia, Ophrys straussii, Anacamptis pyramidalis*	*Rhizoctonia* sp., *Aspergillus* sp., *Alternaria* sp., *Penicillium* sp., *Trichoderma* sp. and *Fusarium* sp. were isolated and characterized.	Morphological and microscopic features of the fungi.	Çiğ et al., 2018
Seventeen *Chiloglottis* spp.	*Tulasnella prima* and *Tulasnella sphagneti* were isolated.	Identified through sequences analysis of the ITS nuclear rDNA and Simple Sequence Repeat loci (SSR) genotyping and population genetic analyses.	Ruibal et al., 2017
Dendrobium friedericksianum	Six endophytic fungi were isolated in which three isolates were binucleate *Rhizoctonia*-like fungi. The teleomorphic states of these fungi were morphologically identified as *Tulasnella violea, Epulorhiza repens* (anamorph *Tulasnella*) and *Trichosporiella multisporum*. The other three species were saprophytic endophytes namely, *Fusarium* (two species) and *Beauvaria*.	Based on cultural characteristics (colorless colonies, right-angle branching, constrictions at branch point, septum in the branch hyphae near its point of origin, chains of monilioid cells, sclerotia formation, and number of nuclei in young cells) the isolates were identified as *Rhizoctonia*-like orchid endophytes. The other isolates were identified based on their macroconidian and conidiogenous cells morphology.	Khamchatra et al., 2016b
Chloraea riojana	Twenty-one root associated fungi including *Alternaria*-like (6 isolates), Penicillium (3), *Trichoderma* (3), unidentified ascomycetous (6) and *Rhizoctonia*-like morphology (3) were recovered. *Rhizoctonia*-like orchid mycorrhizal symbionts were *Rhizoctonia solani/ Thanatephorus cucumeris*.	Based on sequences analysis of the ITS of nuclear rDNA.	Fracchia et al., 2016
Paphiopedilum spicerianum	Some OMF such as *Tulasnella, Ceratobasidium, Russula, Mycena* and a number of unclassified fungi such as *Tricholoma, Fusarium Gibberella, Mortierella Tricholoma Clonostachys, Cylindrocarpon, Gliocladiopsis, Tomentella* etc. were reported.	Next generation sequencing by Illumina MiSeq with primer pair ITS-3 and ITS-4 of amplified ITS2 region was conducted to identify and explore the associated unculturable fungal community structure and dynamics through seasonal and environmental changes.	Han et al., 2016
Paphiopedilum villosum	Three OMF namely, *Tulasnella* sp., *Ceratobasidiumsp.*, and *Flavodon* sp., were identified.	Both morphological and molecular approaches were used for identification.	Khamchatra et al., 2016a

(*continued*)

Table 1 (*Continued*)

Orchid species	Associated Fungi	Method of identification	Refs.
Ophrys bertolonii	An OMF namely, *Epulorhiza* sp. (anamorphic stage of *Tulasnella*) and some ascomycetes fungi namely, *Fusarium, Gibberella* (anamorph *Fusarium*), and *Alternaria* were identified.	Based on sequences analysis of the ITS of nuclear rDNA.	Pecoraro et al., 2015
Pheladenia deformis	OMF such as *Sebacina* sp. was isolated and identified.	Based on ITS sequences analysis of nuclear rDNA of the widespread fungal operational taxonomic units (OTUs).	Davis et al., 2015
Appendiculata sp., *Bulbophyllum beccarii* and *Calanthe vestita*	OMF such as *Ceratorhiza* sp. *Ephuloriza* sp. and *Tullasnela* sp. were isolated and identified.	Identified the fungal endophytes based on morphological characteristics (such as the colony color, concentric circles, hyphal cell size, number of nuclei per vegetative and monilioid cell, sclerotia), observation of peloton in root tissue and grouping of isolates to study ananstomosis ability.	Suryantini et al., 2015
Phalaenopsis (*Phalaenopsis* Taisuco Snow × *Doritaenopsis* White Wonder)	Twenty one fungal species including two common orchid mycorrhizal symbionts namely, *Tulasnella* (anamorphic *Epulorhiza*) and *Ceratobasidium* were isolated and identified.	Metagenomics analysis using species-specific barcoding primers with next-generation sequencing (NGS) of four nuclear ribosomal markers, two nrITS regions (ITS1/2 and ITS3/4), two in the nrLSU region (nrLSU-LR and nrLSUU), the large subunit of the mitochondria ribosomal region (mtLSU) and the sixth subunit of mitochondrial ATPase (mtATP6).	Huang et al., 2014
Microtis media	Four potential OMF manely, *Piriformospora indica, Sebacina vermifera, Tulasnella calospora* and *Ceratobasidium* sp. were isolated and identified.	Sequencing and analysis of nrLSU region of the ribosomal DNA	De Long et al., 2013
Vanilla sp.	*Thanatephorus* spp was isolated and identified.	Nuclear ribosomal ITS region sequencing and analysis	Otero et al., 2013
Aerides multiflorum and *Rhynchostylis retusa*	*Ceratobasidium* sp. AM and *Ceratobasidium* sp. RR	Cultural, microscopic features and ITS region analysis of nuclear ribosomal DNA	Hossain et al., 2013
Piperia yadonii	Fungi belong to the family Ceratobasidiaceae, Sebacinaceae and Tulasnellaceae	Nuclear ribosomal ITS region analysis	Pandey et al., 2013
Holcoglossum rupestre and *H. flavescens*	Isolated 46 culturable fungal endophytes belonging to four classes, *i.e.*, Sordariomycetes, Dothideomycetes, Agaricomycetes, Leotiomycetes. Among the fungal isolates, only two, *i.e. Tulasnella calospora* and *Epulorhiza* spp. were orchid mycorrhizal fungi.	Based on sequences analysis of the ITS nuclear rDNA region.	Tan et al., 2012
Bulbophyllum neilgherrense and *Vanda testacea*	Twenty anamorphic fungal taxa belonging to *Alternaria, Aspergillus, Gliocladium, Paeciliomyces, Phialophora*, and *Penicillium* sp. were isolated and identified.	Identified the fungi based on morphological features.	Sudheep and Sridhar, 2012
Paphiopedilum thailandense	A new orchid mycorrhizal fungus, 'KMI' (KMI refers to Kyoto-Ma- tsubara-Ishii).	Based on sequences analysis of the ITS nuclear rDNA region and protein analysis	Matsubara et al., 2012
Hexalectris warnockii, H. grandiflora, H. brevicaulis, H. revoluta, H. parviflora, H. arizonica, H. colemanii, H. nitida, and *H. spicata*	A number of fungi belonging to Thelephoraceae, Russulaceae and Sebacinaceae families were isolated and identified.	Sequencing of ITS region of nuclear rDNA and phylogenetic analysis.	Kennedy et al., 2011
Changnienia amoena	18 fungal isolates belonging to Ascomycetes, Zygomycetes and Basidiomycetes were isolated and identified.	Sequencing of ITS region of nuclear rDNA and phylogenetic analysis.	Jiang et al., 2011
Dendrobium officinale	*Mycena* sp. was isolated and identified.	Morphological, cultural and sporulation characteristics through light microscopy and SEM; and sequence analysis of the ITS nuclear rDNA were applied.	Zhang et al., 2012
Bipinnula fimbriata	Seven isolates were previously identified as the form-genus *Rhizoctonia* belonging to three groups, *Ceratobasidium* sp., *Tulasnella calospora* and *Thanatephorus cucumeris*.	Morphological (cultural characters, hyphal morphology, nuclear number per vegetative and monilioid cells etc.) and molecular characterization through sequencing of ITS region of nuclear rDNA and phylogenetic analysis.	Steinfort et al., 2010
Dendrobium loddigesii	*Fusarium, Acremonium, Chaetomella, Cladosporium, Colletotrichum, Nigrospora, Pyrenochaeta, Sirodesmium*, and *Thielavia* were isolated and identified.	Used sequencing of 5.8S ITS regions rDNA and phylogenetic analysis for identification of fungi.	Chen et al., 2010
Gastrodia confusa	*Mycena* sp. were isolated and identified.		Tsujita et al., 2009

(*continued*)

Table 1 (*Continued*)

Orchid species	Associated Fungi	Method of identification	Refs.
		Applied the sequencing of ITS region and large subunit nrDNA sequences and phylogenetic analysis.	
Cymbidium goeringii and *C. faberi*	35 species of fungi isolated	Sequencing of ITS region of nuclear rDNA and phylogenetic analysis.	Wu et al., 2009
Dendrobium nobile	*Xylaria* spp., *Guignardia mangiferae*, *Clonostachys rosea, Trichoderma chlorosporum, Fusarium solani* etc. were isolated and identified.	Identified by SEM, sequencing of ITS region of nuclear rDNA and phylogenetic analysis.	Yuan et al., 2009
Cremastra appendiculata, Pleione bulbocodioides, Pleione yunnanensis	*Ceratorhiza* sp., *Epulorhiza* sp., Moniliopsis sp. were isolated from *Cremastra appendiculata, Ceratorhiza* sp., *Epulorhiza* sp. were from *Pleione bulbocodioides* and *Ceratorhiza* sp., *Epulorhiza* sp. were isolated from *Pleione yunnanensis*.	Fungi were isolated from pelotons incubated in sterile distilled water with antibiotics and identified based on cultural morphology and microscopic features.	Zhu et al., 2008
Epipactis helleborine	Several ectomycorrhizal taxa of the Pezizales, such as *Wilcoxina,Tuber, Hydnotrya, Helvella, Genea* and *Exophiala*. In a few individuals of *Leptodontidium, Ceratobasidium, Cenoccocum, Russula, Nectria,* and *Trichoderma* were also found.	Used molecular phylogenetic analysis of ITS and large subunit (LSU) of rDNA sequences.	Tsujita and Yukawa, 2008
Epipogium roseum	*Coprinellus disseminatus* (≡ *Coprinus disseminatus*) was isolated and identified.	Identified based on morphological and anatomical characteristics of the basidiomata.	Yagame et al., 2008
Cephalanthera falcata	Ectomycorrhizal fungi belong to Thelephoraceae or Russulaceae. The identified fungi were closely related to *Tomentella* spp. or *Thelephora* spp.	Sequence analysis of the ITS region of nuclear rDNA.	Yamato and Iwase, 2008
Dactylorchis purpurella	*Rhizoctonia repens* and *R. solani* were isolated and identified.	Characterized by cultural morphology and microscopic features.	Stewart and Kane, 2006
Ionopsis utricularioides and *Tolumnia variegata*	*Ceratobasidium* spp. was identified.	Sequencing of ITS region of nuclear rDNA and phylogenetic analysis	Otero et al., 2004, 2007
Vanilla planifolia, V. poitaei, V. phaentha, and *V. aphylla*	*Ceratobasidium, Thanatephorus* and *Tulasnella* were identified.	Sequencing of ITS region of nuclear rDNA and part of the mitochondrial ribosomal large subunit (mtLSU) and phylogenetic analysis	Porras−Alfaro and Bayman, 2007
Goodyera repens	*Ceratobasidium cornigerum* (syn. *Ceratorhiza goodyerae-repentis* and *Rhizoctonia goodyerae-repentis*) was identified.	Molecular analysis of genomic DNA	Cameron et al., 2006
Pterostylis nutans, Pterostylis longifolia, Pterostylis bicolor	*Ceratorhiza* sp. was isolated and identified.	Molecular analysis of genomic DNA	Midgley et al., 2006
Epidendrum rigidum, Isochilus lineares, Maxillaria marginata, Oncidium flexuosum, O. varicosum, Oeceoclades maculata, Polystachya concreta	Seven fungal isolates belong to *Ceratorhiza* sp., *Epulorhiza epiphytica* and *Epulorhiza repens*	A combination of morphological and molecular characterizations techniques including RAPD, PCR−RFLP and ITS sequence analysis of rDNA	Pereira et al., 2005
Epipogium roseum	*Coprinus* spp. or *Psathyrella* spp. were isolated and identified.	Phylogenetic analysis based on the ITS sequences of nuclear rDNA	Yamato et al., 2005; Yagame et al., 2006
Tolumnia variegata	*Thanatephorus cucumeris* (=*Rhizoctonia solani*), *Ceratobasidium* spp. were isolated and identified.	Analysis of nuclear ribosomal ITS region.	Otero et al., 2005
Cypripedium calceolus, C. candidum, C. californicum, C. fasciculatum, C. guttatum, C. montanum and *C. parviflorum*	Members of the fungal family Tulasnellaceae. Rarely occurring root endophytes include members of the Sebacinaceae, ceratobasidiaceae, and the ascomycetous genus, *C. californicum* and *Phialophora* sp. were isolated and identified.	Through analysis of nuclear large subunit and mitochondrial large subunit sequence and RFLP (restriction fragment length polymorphism)	Shefferson et al., 2005
Cypripedium fasciculatum S. Watson	Several fungal species, at least one of which belongs to the family Russulaceae *i.e. Russula* and a known OMF *Tulasnella* were isolated and identified.	Identified by analysis of fungal DNA with PCR-RFLP and Phylogenetic analysis through ITS sequences.	Whitridge and Southworth, 2005
Tolumnia variegate and *Ionopsis utricularioides*	Most of the mycorrhizal isolates were four different clades of *Ceratobasidium*.	Using phylogenetic analysis through ITS sequences, and symbiotic seed germination experiments.	Otero et al., 2004
Spathoglottis plicata	Three genera and four species of mycorrhizal fungi were identified namely *Epulorhiza repens, Rhizoctonia globularis, Rhizoctonia* sp. and *Sabacina* sp. (*Epulorhiza* sp. anamorph).	Identification was based on morphological characteristics (colony growth pattern, growth rate, color, sclerotia formation etc.).	Athipunyakom et al., 2004
Cephalanthera austinae, Goodyera pubescens, Liparis lilifolia, Tipularia discolor	*Tulasnella* spp. and *Sebacina* spp. were isolated and identified.	ITS and mtLSU sequence analyses	McCormick et al., 2004
Epidendrum rigidum and *Polystachia concreta*	*Epulorhiza epiphytica* sp. nov. was isolated and identified.	By cultural morphology and microscopic features.	Pereira et al., 2003

(*continued*)

Table 1 (*Continued*)

Orchid species	Associated Fungi	Method of identification	Refs.
Hexalectris spicata var. *spicata, H. spicata* var. *arizonica* and *H. revoluta*	*Thanatephorus ochraceus* (Ceratobasidiaceae) was an occasional colonizer of mycorrhizal roots and nonmycorrhizal rhizomes. Members of the Sebacinaceae were the primary mycorrhizal fungi in every root and were phylogenetically intermixed with ectomycorrhizal taxa.	Identified based on RFLP and sequences analysis of the ITS and nuclear LSU ribosomal gene fragments and microscopic features.	Taylor et al., 2003
Campylocentrum fasciola, Campylocentrum filiforme, Erythrodes plantaginea, Ionopsis satyrioides, Ionopsis utricularioides, Oeceoclades maculata, Oncidium altissimum, Psychilis monensis, and *Tolumnia variegata*	Isolated 108 *Rhizoctonia*-like fungi include anamorphs of *Tulasnella, Ceratobasidium*, and *Thanatephorus* and all endophytes were closely related to different anastomosis groups of *Ceratobasidium* spp. were isolated and identified.	Identified by ITS region sequence analysis of genomic DNA.	Otero et al., 2002
Eulophia flava, Goodyera procera, Habenaria dentata, and *Spiranthes hongkongensis*	*Ceratorrhiza cornigerum, C. globisporum, Rhizoctonia repens, R. solani*, and *Thanatephorus cucumeris* were isolated and identified.	Through cultural and microscopic features of the fungal isolates such as TEM, RAPD, RFLP and CAPS analysis of ITS of rDNA.	Shan et al., 2002
Cymbidium goeringii	Three species of *Rhizoctonia* namely, *R. repens* (anamorph state of *Tulsanella repens*), *R. endophytica* (*Ceratobasidium cornigerum*), and an unidentified species (possibly an anamorph of *T. calospora*). were isolated and identified.	By Analysing of 18 s rDNA sequences.	Lee, 2002
Dactylorhiza majalis	*Tulasnella* and *Laccaria*	Through analysis of the mitochondrial ribosomal large subunit (LSU) DNA and Single-strand conformation polymorphism (SSCP).	Kristiansen et al., 2001a
Neuwiedia veratrifolia	*Rhizoctonia* sp.	Microscopic features, cultural morphology and cytological appearance of the endophytes	Kristiansen et al., 2001b
Dactylorhiza maculata	*Rhizoctonia goodyerae-repentis*	Identified based on morphological features.	Kulikov and Filippov, 2001
Platanthera bifolia, Gymnadenia conopsea, Malaxis monophyllos	*Rhizoctonia repentis*		
Neottianthe cucullata	*R. goodyerae-repentis*		
Goodyera repens	*R. goodyerae-repentis*		
Coeloglossum viride, Cypripedium calceolus	*Rhizoctonia* sp.		
Dactylorhiza majalis	*Tulasnella deliquiscens, Tulasnella calospora, Epulorhiza anaticula*, and *Tulasnella violea*	Through analysis of polymorphism in mitochondrial ribosomal large subunit DNA sequences.	Andersen 1990; Kristiansen et al., 2001a,2001b
Pterostylis accuminata	*Rhizoctonia solani*	Molecular analysis of genomic DNA	Pope and Carter, 2001
Corallorhiza maculate and *C. mertensiana*	Fungi belonged to the family Russulaceae	By ITS restriction fragment length polymorphisms (RFLPs) analysis.	Taylor and Bruns, 1999
Platanthera clavellata, P. cristata, P. integrilabia, P. ciliaris	*Epulorhiza inquilina*	Molecular analysis of genomic DNA	Currah. et al., 1997b; Zettler and Hofer, 1998
Platanthera obtusata	*Ceratorhiza goodyerae-repentis*	Molecular analysis of genomic DNA	Currah and Sherburne, 1992
Coeloglossum viride	*Moniliopsis anomala*	Molecular analysis of genomic DNA	
Calypso bulbosa	*Epulorhiza anaticula*		
Platanthera obtusata	*Ceratorhiza goodyerae-repentis*		
Dendrobium dicuphum	*Tulasnella irregularis*	Molecular analysis of genomic DNA	Warcup and Talbot, 1980

among them are also suggested in this regard. It is demonstrated that a single fungus can develop mycorrhizal association with more than one orchid, and similarly, an orchid can harbor more than one fungal endophyte (Jacquemyn et al., 2014, 2015; Li et al., 2022). For example, Porras–Alfaro and Bayman (2007) found that *Tulasnella* and *Ceratobasidium* were present in both *Vanilla poitaei* and *V. planifolia*. Investigations among 134 individuals of *Hexalectris* species from 42 populations revealed that *Hexalectris warnockii* had associated solely with the fungi belong to Thelephoraceae. *H. grandiflora* and *H. brevicaulis* were associated exclusively with the Sebacinaceae and Russulaceae, while *H. spicata* species complex was associated largely with unique fungal sets belong to Sebacinaceae (Kennedy et al., 2011). Another investigation among 44 orchid species of different life forms observed that some orchids (*e.g. Epigeneium amplum, Nervilia plicata* and *Oberonia variabilis*) associated with only a single fungal taxon so as to corroborated extreme host specialization while the majority of the orchid species were networked with several fungal partners at the same time, indicating the extreme host specialization cannot be the sole explanation (Xing et al., 2019). Taylor et al. (2002) claimed that specificity symbolizes a continuum that can be properly evaluated by utilizing phylogenetic information only. They summarized a number of inconsistent results of mycorrhizal specificity in their investigations with a mega list of mycorrhizal fungi and phylogenetic data. However, based on available information they commented that with a few exceptions all green orchids have little or no specificity while fully myco-heterotrophic orchids have species specificity. An extensive investigation was made by Shefferson et al. (2005) to assess species–specificity of mycorrhizal fungus in Lady's slipper orchids (*Cypripedium* spp.). They collected 59 plants of seven *Cypripedium* spp. from 44 different populations in Europe and North America and identified their associated fungal endosymbionts and reported that *Cypripedium* mycorrhizal fungi were mainly the members of family

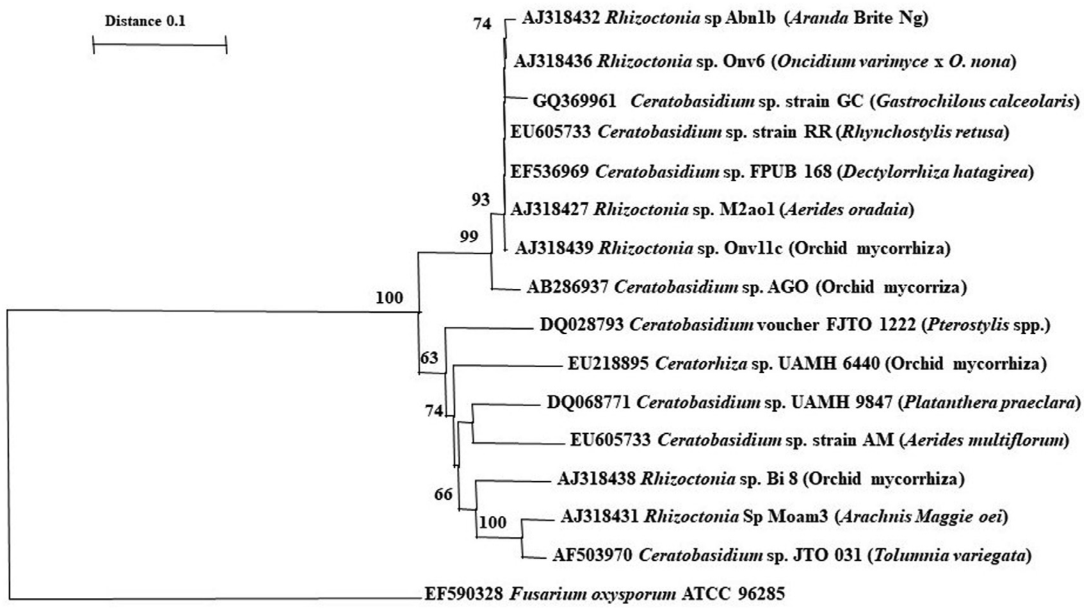

Fig. 3. A phylogenetic tree showing relationship among some orchid mycorrhizal fungi based on ITS region analysis. Numbers on the nodes indicated bootstrap values. *Fusarium oxysporum* ATCC 96,285 was used as the out-group. Bar, 0.1 substitution per site. (unpublished photographs of author).

Tulasnellaceae. Several root endophytes were affiliated to Ceratobasidiaceae and Sebacinaceae. A single Ascomycetous genus, *Phialophora* was also present. They also noted that *C. californicum* was a unique species with superficially low specificity because it was associated with Ceratobasidioid, Sebacinoid, and Tulasnelloid fungi in apparently equal proportions. Based on these observations they hypothesized that high specificity is not limited to non-photosynthetic orchids only. Orchid seedlings that germinated naturally associated with a broader range of fungi (Zelmer et al., 1996). This might be a point towards a change of mycobionts during the development of the individual plant; however, it might also reflect a fairly unspecific germination with subsequent decimation of seedlings, leaving only those with most favorable symbiosis to develop into adult plants. In the case of *Gastrodia elata*, however, a shift in mycobiont appears to be the rule, from *Mycena osmundicola* to *Armillaria mellea* (Xu and Mu, 1990; Chen et al., 2019, Li et al., 2022).

8. Mycorrhiza and conservation of orchids

Orchidaceae is the most threatened family of flowering plants. Taking into consideration the present status of orchids, the entire family was included in the Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). Various human activities, including land clearing for agriculture, deforestation, brutal collection by orchid lovers, superfluous use of orchids for ethno-medicinal purposes, destruction of orchid habitats for urbanization, mining, killing pollinators, shifting cultivation, weed invasion, grazing, altered environmental conditions and illegal trade has led to decrease the wild populations of orchids. Meanwhile a large number of orchids have been extinct and many orchids are now fall in rare and endangered condition. Therefore, it is utmost necessary to take efficient conservation measures for sustainable management of orchids and meet up our increasing demand.

It is well recognized that all orchids show mycoheterotrophy in their early developmental stages and mostly the relationship persists all over their lifespan. The chlorophyllous orchids could reduce their fungal dependency once they achieved the photosynthetic capability but achlorophyllous orchids are entirely reliant on mycorrhizal fungi all over their lifespan for both energy and nutrient requirements. It is apparent that, in natural habitats all epiphytic and numerous mature terrestrial green orchids are less dependent on direct sunlight for energy, it is due to the presence of mycorrhizal fungi, however it needs additional investigation. OMF have been acknowledged as a significant determining factor for survival of an orchid to its habitat. The interactions between mycorrhizal fungi and orchids have been studied extensively in past two decades, and concluded that mycorrhizal fungi have the most important roles in the conservation and management of orchid species. Therefore, habitat protection could be a critical approach to orchid conservation.

9. Application of OMF for bio-hardening of TC raised seedlings

Hardening of tissue culture (TC) raised plantlets by compatible fungi is another important aspect of orchid mycorrhiza. It is of theoretical interest and practical importance to horticulture. The *in vitro* produced plantlets are rigorously encountered by major and minor microbial pathogens and abiotic stresses when transfer to outside environment from culture room and cannot develop resistance against these factors as they are not absolute autotroph. Additionally, physiological and anatomical deficiencies of tissue culture raised plantlets such as poor cuticle layer in leaves, reduced photosynthetic ability, malfunctioning stomata, and a few hairs or lack of hairs in roots make them incompetent to survive under adverse environmental conditions. Since orchid seeds germinate in nature with the help of mycorrhizal fungi only and the fungi reside in the roots of growing seedlings, supply nutrition and develop defense mechanisms against abiotic and biotic factors and prolong their occurrence to adult orchids; there is a prospect of bio-hardening for enhancing survival and growth of *in vitro* grown seedlings by co-cultivating with compatible mycorrhizal fungi. Asymbiotically raised plantlets of *Cephalanthera falcata* were successfully re-introduced into the natural environment with OMF (Yamato and Iwase, 2008). Fungal colonization was observed in the transplanted *C. falcata* seedlings as well as wild plants, and the associated fungi were characterized belong to the fungal family Russulaceae or Thelephoraceae, used to develop ectomycorrhiza with its host trees. Therefore, a tripartite relationship among mycorrhizal fungi, *C. falcata*, and host trees was developed. This tripartite relationship was confirmed through investigation of ^{14}C movement among *Corallorhiza trifida* (orchid), *Ceratobasidium cornigerum* (fungus) and *Betula pendula* or *Salix repens* (autotrophic shrub) (McKendrick et al., 2000). The tripartite relationship was further confirmed among *Rhizanthella gardneri* (a unique subterranean orchid species entirely grown under soil even during flowering), a fungus (*Ceratobasidium* sp., associated with *R. gardneri*) and *Melaleuca scalene* (an autotrophic shrub) through exploration of isotopically labeled tracers, double-labeled [^{13}C-^{15}N] glycine and $^{13}CO_2$ (Bougoure et al., 2010). A tripartite harmonization of nutrient sharing among mycorrhizal fungi, orchids, and the adjacent host trees possibly be more general than the association developed only between orchids and their fungal symbionts. Therefore, there is another possibility to use fungi for introducing artificially propagated orchids into natural habitats. *In vitro* raised plantlets of a hybrid *Cymbidium* were efficiently hardened by six different strains of mycorrhizal fungi, collected from different wild terrestrial orchids and the average increment of fresh weight of inoculated seedlings was higher as compared to control (Fang et al., 2008). *In vitro* raised plantlets of *Anoectochilus roxburghii* formed endotrophic mycorrhiza after co-cultivation for 60 days with an *Epulorhiza* sp. (AR-18) and markedly promoted growth of this orchid by the fungus (Li et al., 2012). Noteworthy increases in leaf size and fresh weight of *Anoectochilus formosanus* and *Phalaenopsis* was observed when the plantlets were treated with *Ceratobasidium* sp. AG-A and *Ceratobasidium* sp. AG-A (R02) respectively (Cheng and Chang, 2009; Wu et al., 2011). The *Ceratobasidium* AR2 has prodigious potential as a growth stimulating fungus for *A. roxburghii* cultivation because it significantly increased average plant height, number of buds per plant, number of roots, fresh weight, and dry weight, as compared with those of control (Zhang et al., 2020a). *In vitro* raised seedlings of *Cymbidium aloifolium* and *C. giganteum* exhibited 100% survival when inoculated with a mycorrhizal fungus *Ceratobasidium* RR characterized from *Rhynchostylis retusa* (Hossain et al., 2013). Nowadays many scientists are trying to devise an efficient technique for growth promotion and improve survival rate after transplantation of the seedlings to natural environment through developing symbiotic associations with orchids and compatible fungi (Ke et al., 2008; Zhang et al., 2012; Ye et al., 2020). Meanwhile, some critical investigations stated that mycorrhizal fungi can efficiently enhance seed germination, *ex vitro* survival, growth, and development of the seedlings (Chang and Chou, 2001; Chang, 2007, 2008; Kang et al., 2007; Jin et al., 2009; Ye et al., 2020) (Table 2). For instance, growth of seedlings of *Acampe praemorsa, Anoectochilus formosanus, A. roxburghii, Bletilla formosana, Cymbidium aloifolium, C. giganteum, C. goeringii,* C. *kanran, Dendrobium amethystoglossum, D.* Snow Flake, *D. officinale, D.* Chao Praya Garnet, *D. tobaense, Doritis pulcherrima, Doritaenopsis, Haemaria discolor* var. *dawsoniana, Oncidium* Gower Ramsey, *Paphiopedium* spp., and *Phalaenopsis* were significantly enhanced by particular *Rhizoctonia* and *Ceratobasidium* strains (Guo et al., 2000; Chang and Chou, 2001; Ke et al., 2008; Wu et al., 2010; Zhang et al., 2012; Hossain et al., 2013; Wu et al., 2013; Li et al., 2012; Sathiyadash et al., 2013; Zhang et al., 2020a, 2020b; Ye et al., 2020). The survival rate of *in vitro* raised seedlings of *D. officinale* was achieved 80∼90% after transfer to outside environment with specific OMF (Zhang et al., 2012). OMF manufacture some important bioactive compounds / growth stimulus such as indole-3-acetic acid (IAA), α-naphthalene acetic acid (NAA), heteroauxin, dormin, zeatin, zeatin riboside and gibberellic acid (GA_3) that stimulate the growth and development of *Dendrobium nobile, D. candidum D. officinale*, and *Pleione bulbocodioides* saplings (Dan et al., 2012b; Sathiyadash et al., 2020). NAA and GA_3 produced by OMF promoted root and stem elongation of *D. huoshanense* (Zhang et al., 1999). Seven fungus-responsive genes were identified in *Anoectochilus roxburghii* inoculated with *Epulorhiza* sp. (AR-18) of which four genes (*AR-DD017, AR-DD019, AR-DD020*, and *AR-DD023*) were directly involved in promoting growth and development of seedlings (Li et al., 2012). The *AR-DD017* gene encodes a hypothetical protein partially identical to some fungal and animal proteins but the exact role of this protein in plant growth and development was undefined. The *AR-DD019* and *AR-DD020* gene encode uracil phosphoribosyltransferase (UPRTs) and an amino acid transmembrane transporter, respectively. The *AR-DD023* gene was specifically functioned in *A. roxburghii* only in the inoculated seedlings by encoding a maturase K (*matK*) and a tRNA-Lys (*trnK*). UPRTs are engaged in saving pyrimidines through catalyzing the manufacture of uridine monophosphate from phosphoribosyl-pyrophosphate and uracil and were labeled as energy-saving enzymes. Now it is experimentally demonstrated that pyrimidines mainly uracil salvage have crucial influence on plant growth and development (Mainguet et al., 2009). As a result of the improvement of molecular biology, and systems biology, it is now possible to study precisely the orchid-fungal interaction through metabolome and transcriptome analyses. Root transcriptome analysis of *Cymbidium hybridum* co-cultivated with diverse symbiotic fungi, it was detected that reactive oxygen species detoxification, cell wall modification, phosphate transport, and defense-related plant hormones were co-induced in all symbiotic associations (Zhao et al., 2014). Through a combination of quantitative real time reverse transcriptase polymerase chain reaction (qRT-PCR), next generation RNA sequencing, and gas chromatography it was confirmed that *Mycena* sp. MF23, a mycorrhizal fungus stirred the accumulation of dendrobine in *Dendrobium nobile* stems through regulating gene expression engaged in the mevalonate pathway (Li et al., 2017). Transcriptome analyses of differentially expressed genes (DEGs) of the control and *Ceratobasidium* sp. AR2 inoculated *Anoectochilus roxburghii* revealed that mycorrhizal fungi involved in different metabolic pathways and secondary metabolites biosynthesis (Zhang et al., 2020a). Quantitative real time PCR and Liquid chromatography−mass spectrometry showed that *Ceratobasidium* sp. AR2 induced flavonoids accumulation by regulating gene expression related to flavonoid biosynthesis pathway (Zhang et al., 2020b). These results provided molecular groundwork of the consequence of mycorrhizal fungi on flavonoid biosynthesis in *A. roxburghii* and endow with novel acuities to improve the quality and yield of orchids.

Resistance to pathogenic diseases was stimulated and reduced disease severity by mycorrhizal fungi in many ornamental orchids, for example *Cymbidium* spp., *Doritaenopsis* spp., *Oncidium* sp., *Phalaenopsis* spp., *Phaphiopedium delenatii* etc. (Chang 2007, 2008; Fang et al., 2008; Ke et al., 2008). *Phalaenopsis* plantlets inoculated by *Rhizoctonia solani* as well as several strains of *Ceratobasidium* sp. exhibited reduction of the severity of soft rot disease; a correlation was also obvious between reduction of disease symptom and plant growth (Wu et al., 2011). Similarly, *Vanilla* plants colonized with *Ceratobasidium* were shown to control *Fusarium* originate root rot disease (Bayman et al., 2011). Therefore, OMF can be applied for (i) increase *ex vitro* survival rates of *in vitro* raised plantlets, (ii) reduce disease infection rates, (iii) improve both reproductive and vegetative growth of plants, (iv) induce early flowering and quality improvement of flowers, and v) reintroduction of orchids in natural habitats for conservation and sustainable management.

10. Prospects of OMF research

Patterns of fungal specificity in orchids were studied in terms of their adaptive and evolutionary consequences as well as their role in orchid speciation. Better understanding the biology of mycorrhizal associations of rare and endangered orchids is also essential for conservation measures (Dearnaley et al., 2012; Ruibal et al., 2017). There is a bright potential of application of OMF in commercial cultivation of orchids by reducing the usage of environmentally harmful chemical inputs *i.e.* pesticides and fertilizers; because mycorrhizal fungi are eco-friendly, have vital role in nutrient uptake by plants and improve flexibility to environmental stresses (Suz et al., 2018). Methods for commercial production of AM fungi have been developed and their application in agriculture and horticulture demonstrated inexpensive, more dependable and led to significant yield increase. However,

Table 2
Application of OMF for growth promotion, survival and reduction of disease severity in the orchid seedlings.

Orchids species	Fungal partner	Influence/role of OMF	Refs.
Dendrobium officinale	*Tulasnella* spp.	The tested OMF showed a strong ability to promote seedling growth of *D. officinale*. Based on the experimental outputs it was suggested that *in situ* seedling baiting technique is an effective way to identify seedling growth-promoting fungi for practical use in restoration plantings, and the method have broad applications in orchid mycorrhiza studies and orchid conservation.	Chen et al., 2021
Anoectochilus roxburghii	*Ceratobasidium* sp. AR2	This observation indicated that mycorrhizal fungi could significantly promote the growth and development of orchid plants. The findings also highlighted the molecular basis of the effects of mycorrhizal fungi on biosynthesis and accumulation of flavonoids which is a novel insight to improve the yield and quality of this orchid.	Zhang et al., 2020a
Dendrobium officinale	6 fungal strains (4 *Tulasnella* strains and 2 *Sebacina* strains)	The fungal strains could promote seed germination and seedling growth of *D. officinale* with different efficiencies. The *Tulasnella* strains S6 and S7 conferred higher germination percentage and rapid protocorm development compare to other fungal strains. The seedlings inoculated with *Sebacina* strain S3 showed highest fresh and dry matter yield followed by *Tulasnella* strains S6 and S7. Seedling inoculated with *Sebacina* strain S2 greatly enhanced root and tiller production and the content of total crude polysaccharides.	Zhang et al., 2020b
Catasetum tabulare and two hybrid orchids (*Miltoniopsis Kumbia* and LC. Irene Finney)	Five macromycetes fungi namely, *Pleurotus ostreatus Var. Columbinus, Pleurotus citrinopileatus, Lentínula edodes, Laetiporus sulphureus* and *Agaricus bisporus*	The outcome of the research proved an improved survival rates and enhanced development of the co-cultured plantlets in comparison to the control.	Manuel and Jeannette, 2020
Dendrobium officinale	*Sebacina* strains S2 and S3	Seedlings inoculated with *Sebacina* strains S2 and S3 showed optimal crude polysaccharides content and fresh and dry matter yield.	Zhang et al., 2020b
Oncidium hybrid	*Piriformospora indica* (an endophytic fungus of Sebacinales)	*Piriformospora indica* colonizes *Oncidium* orchid roots and promotes growth and plant performance was Confirmed by expression analysis of miRNAs and target genes. In this study it was demonstrated that after 8 weeks of co-cultivation with the fungus the fresh weight of seedlings was significantly increased, which includes a higher leaf number, an increase in the root number and root diameter bigger stem diameter, and longer height of the seedlings.	Ye et al., 2014
Acampe praemorsa		The *in vitro* raised seedlings inoculated with OMF survived under *ex vitro* conditions	Sathiyadash et al., 2013
Cymbidium kanran Cymbidium goeringii Cymbidium sp.	*Rhizoctonia* sp.	Colonization of roots by OMF increases plant height, biomass, root formation, and root length	Lee et al., 2003; Wu et al., 2013, 2010
Dendrobium nobile, D. loddigesii Dendrobium officinale Haemaria discolor Anoectochilus formosanus	*Tulasnella* repens and other OMF	OMF enhanced the plant biomass	Zaiqi and Yin, 2008; Yang et al., 2008; Chang and Chou, 2001, 2007
Anoectochilus roxburghii Dendrobium candidum and *D. nobile*	*Epulorhiza* sp., *Mycena dendrobii, Moniliopsis* sp., *Gliocladium* sp., *Mycena anoectochila*	OMF colonized plants exhibited taller and had higher biomass	Dan et al., 2012a, 2012b
Cymbidium	Six strains of unidentified OMF endophytes	Plant colonized by OMF showed higher biomass than un-inoculated seedlings	Fang et al., 2008
Doritaenopsis and *Phalaenopsis*	*Rhizoctonia* and *Ceratobasidium*	Seedlings colonized by OMF had higher biomass than un-inoculated one.	Wu et al., 2009, 2011

(*continued*)

Table 2 (*Continued*)

Orchids species	Fungal partner	Influence/role of OMF	Refs.
Guarianthe skinneri	*Trichoderma harzianum*	The *in vitro* raised seedlings acclimatized by fungi showed taller with more leaf and shoot numbers	Gutierrez-Miceli et al., 2008
Cattleya aurantiaca and *Brassavola nodasa*	*Epulorhiza*	Seedlings colonized by OMF were taller and heavier than their non-mycorrhizal counterparts	Ovando et al., 2005
Dendrobium officinale	*Mycena* sp.,	Roots colonization by OMF increased plant height, biomass, and the number of new buds	Zhang et al., 2012
Doritaenopsis Taisuco Wonder 'King Car Butterfly KC1111' and *Phalaenopsis* Tai Lin Redangel 'V31'	*Ceratobasidium* sp. AG-A (R02) and *Rizoctonia solani* AG-6 (R04)	Inoculated with OMF strains significantly increased the growth and reduced the severity of soft rot (by *Erwinia* spp.), and a correlation also exists between disease symptom reduction and plant growth	Wu et al., 2011
Vanilla	*Ceratobasidium*	Colonization by OMF was shown to control the root rot caused by *Fusarium*	Bayman et al., 2011
Spiranthes magnicamporum	Orchid mycorrhiza	Found 100% survival for all the symbiotic seedlings of upon their transfer to soil compared to the 5% survival for the seedlings raised asymbiotically.	Anderson, 1991

enormous potential of OMF remains untapped which may provide an important reserve in future development. The prospect of OMF in propagation and conservation of orchids are (i) OMFs enhance germination of orchid seeds *in vitro* and *ex vitro*, have potential in symbiotic seed germination, particularly for those orchid seeds are recalcitrant in asymbiotic culture, (ii) it increases the survival rates of the tissue culture raised plantlets in outside culture room, so there is a potential of mass scale hardening and establishment in outside environment and conservation of rare, endangered and economically important orchids; however, specificity of orchid-fungus should be considered and compatible fungal strains should be applied, (iii) OMF stimulate nutrient absorption and translocation may be used for better production and management of orchids in commercial firms, (iv) OMF help in earlier flowering and increase flower quality which is very important for floriculture, (v) OMF improve different enzyme activities [superoxide dismutase (SOD), acid phosphatases, alkaline phosphatases etc.], increase chlorophyll contents and ratify antioxidant abilities through increasing quantities of flavonoids, ascorbic acids, polysaccharides and polyphenols in many orchids, (vi) OMF reduce the infection rate through inhibiting the pathogens by competition for nutrients or space and consequently less agrichemicals are required. It is important to mention here that different orchid hosts have some degrees of specificity for OMF. So, before commercial applications of OMF, compatible strains should be selected through random co-cultivation of orchid and fungi in greenhouse or *in vitro*. Growth improvement and disease resistance capacity might be the most unique character of OMF that cannot be easily substituted by other cultural practices.

There are too many evidences that orchids are frequently colonized by various fungi, some of which may not interacts as true mycorrhiza. Advance research on orchid mycorrhizal associations will provide more convenient information to clarify in-depth interaction architecture of mycorrhiza in both autotrophic and achlorophyllous plant-hosts. Since orchids symbolize a vibrant, widespread and diverse plant group, mycorrhizal research is very important for their conservation and sustainable management. The problems of commercial propagation of orchids include high mortality rate of the *in vitro* seedlings and slow growth after transplanting to the field, and difficulties in stimulating blooming in adult plants (Wang et al., 2004). Application of compatible mycorrhizal fungi for better vegetative and reproductive growth created hope to the orchidologists and horticulturists. Asymbiotic seedlings also showed better survival and fitness in outside environment when hardened with mycorrhizal fungi. Some evidence also indicate that orchid mycorrhizal fungi could promote flowering and shorten the time required for first blooming (Zhang and Zhou, 2004). Therefore, isolation, identification and screening of OMF is the primary and very important step for its applications in conservation and commercial cultivation of orchids (Luo et al., 2003; Zhang and Zhou, 2004). Compatible mycorrhizal fungus means that a fungus should have at least one of the following three characters: (1) capable of stimulating seed germination, (2) capability of improving protocorm growth, young saplings or juvenile plantlets, and (3) improving the growth and reproduction of adult plants. It was found that epiphytic orchids hurbour statistically more effective fungi that are able to promote seed germination than terrestrial orchids do (Liu et al., 2010). Once the symbiotic association is developed, the mycorrhizal fungi serve as the leading microorganism to produce antagonistic substances. These substances are effective to control the invasion of harmful pathogenic microorganisms and indirectly improve growth and survival of seedlings (Chen et al., 2003; Zhao and Liu 2008; Jin et al., 2009). Although the mechanism of seedling growth and development by OMF is not crystal-clear, the effects of fungal elicitors, phytohormones, regulation of the function of associated enzymes or the related enzyme gene expression, accumulation of signal molecules and motivation of metabolite pathways were legalized. The use of fungal elicitors to generate symbiotic seedlings for orchid population restorations should be explored further. Some experimental reports thoroughly stated the possibilities of biological control of pathogenic agents by OMF (Otero et al., 2013). Combining luminous way of thinking with competent experimentation it may be recommended the alternative use of OMF for propagation and conservation of economically important orchids.

Available information regarding the impact of OMF is not sufficient on the distribution and survival of orchids in the ecosystem. Knowledge of the distribution of OMF within environment is an important issue to attempt for reintroduction and management of orchid diversity. It will also contribute in better understanding of the reasons of endangered status of certain species of orchids. For example, low fungal diversity and significant host-fungus specificity might restrict orchids to definite habitats. A better understanding of host-fungus specificity will lead successful application of OMF in orchid propagation and conservation in the near future. Mycorrhiza regards as energy and nutrient switch for plants, therefore more extensive studies on mycorrhizal symbiosis are necessary. The ascendancy of the genomic research era on mycorrhizal research put forwards the immense importance of OMF for the years to come. It is also emphasized that a strong interaction among the different disciplines will be essential (Söderström, 2002). In recent decades, significant progress was made in mycorrhizal research through the application of new and advance sophisticated techniques, huge new data were published describing the importance of mycorrhizal fungi and mode of interactions with host and the surrounding environment. However, the great potential of application of mycorrhizal fungi for monitoring the survival ability in the field and reintroduction of orchids are infrequently appeared. Till date, it is not clear that the ecologically-adapted isolates or indigenous fungal strains are suitable for reintroduction and recruitment of orchids in the field conditions. Evidences of current research have also revealed that mycorrhizal fungi are the vital microorganisms to carbon and other nutrient cycling in both natural and agricultural ecosystems. This consciousness is certainly crucial in the times of global environmental changes. Genomics studies of OMF have unlocked new challenges for the investigation of the structure, function and dynamics of orchid mycorrhizal fungi. So, it is utmost important to put attention for the enhancement and improvement of future mycorrhizal research.

Declaration of Competing Interest

The author declares that he has no conflicts of interest.

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